

Instruction format :-

Instruction is a Command given to the Computer to perform a specific operation. It is a group of bits (binary codes) that instructs the Computer to perform specific operation. An instruction format is a binary format which specifies a computer instructions and are divided into groups known as fields. The most common fields are

- 1> UpCode
- 2> Address
- 3> Mode

> An upcode field specifies the operation which is to be executed such as: Add, Sub, Mul, Div etc.

> An address field specifies the process register.

> The mode field specifies the effective address of the operands.

> Computer may have instruction of several different length which contains the number of address. Based on that, there are 3 types of CPU Organization.

- 1> Sign Single Operator Organization

- 2> Register Register

- 3> STACK Organization

Single Acc. Organization :

> All the operations are performed in an implied acc. register with one operand implicitly acc. register which helps to minimize the internal complexity and it allows for a short instruction.